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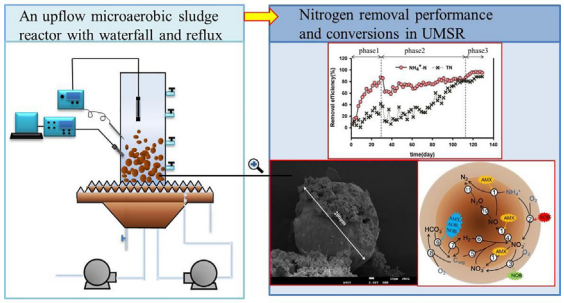
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Efficiency, granulation, and bacterial populations related to pollutant removal in an upflow microaerobic sludge reactor treating wastewater with low COD/TN ratio

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GRAPHICAL ABSTRACT



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ABSTRACT

In this study, a novel upflow microaerobic sludge reactor (UMSR) was constructed to conduct anaerobic digestion of municipal wastewater with low carbon and nitrogen ratio (C/N). Oxygen in the UMSR was supplied by falling water and external recirculation. Excellent nitrogen removal performance was obtained in the UMSR for treating wastewater with low C/N ratio at a temperature of 25 °C and a hydraulic retention time of 24 h. Ammonium and total nitrogen removal efficiencies averaged 92.35% and 90.41%, respectively, and sludge granulation occurred during acclimation. The inferred metabolism of nitrogen removal and ecological positions of functional microbe were integrated into a granule model by scanning electron microscopy. Additionally, the analysis of microbial community indicated that aerobic nitrifying bacteria and heterotrophic bacteria survived on the surface of sludge floc and granules while the anaerobic autotrophic, heterotrophic denitrifying, and anaerobic ammonia oxidation bacteria were present in the inner layer.

1. Introduction

Ammonium and organic material in untreated municipal wastewater are detrimental to the environment of the receiving water (Qin et al., 2015; Wang et al., 2016). Anaerobic digestion process has been widely investigated for treating municipal wastewater because of its low sludge and high energy recovery rate (Maharaj and Elefsiniotis, 2001). However, ammonium left in the anaerobic digestion liquid causes eutrophication (Meng et al., 2015). Therefore, it is necessary to remove nitrogen during the anaerobic digestion of municipal

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wastewater. Conventional nitrogen removal is completed through application of ammonia-oxidizing bacteria (AOB), nitrite-oxidizing bacteria (NOB), and heterotrophic denitrifiers in succession (Guo et al., 2013). The major limitations of conventional nitrification and denitrification processes include high aeration costs, requirement of additional carbon sources for denitrification, and large excess sludge production (Kartal et al., 2010a,b; Laureni et al., 2016).

Microaerobic treatment process is an emerging technology for treating wastewater because of its low dissolved oxygen (DO) demand with concentrations ranging from 0.3 to 1.0 mg/L (Zitomer, 1998). It is also characterized by low yield of excess sludge, high chemical oxygen demand (COD) removal rate, low treatment cost, and small area of occupation (Zheng and Cui, 2012). Additionally, in the microaerobic treatment process, organic substances and nutrients can be removed simultaneously in a single reactor because aerobic bacteria such as AOB, NOB, and heterotrophic aerobic bacteria can propagate at different depths of biological flocs across DO concentration gradients (Zheng and Cui, 2012). Microbial populations, including aerobic bacteria and facultative aerobic bacteria, can grow on the outer layer of the activated sludge floc or granules and consume DO which results in an anaerobic microenvironment in sludge granules or floc and allows anaerobic microorganisms, such as denitrifying bacteria and anaerobic fermentation bacteria, to live in the inner layer (Liu and Dong, 2011). Microaerobic biological technology has been proved to remove nitrogen from domestic wastewater with removal efficiencies of around 80% (Zheng and Cui, 2012). Meng et al., (2018) explored the feasibility of using microaeration conditions for treating piggery wastewater with a low C/N ratio, and the total nitrogen (TN) removal in it achieved above 85%.

Filamentous overgrowth in the microaerobic process causes bulking and loss of sludge. Filamentous bacteria have large specific surface area and strong ability to capture low-concentration organic matter. Therefore, filamentous bacteria survive well under the conditions of low DO and low pollutant concentrations. Granular sludge can effectively relieve sludge bulking and improve settleability of the sludge (Morgenroth et al., 1997). However, the performance of microaerobic processes dominated by granular sludge in anaerobic digestion of municipal wastewater has received less attention.

In the present study, an upflow microaerobic granular sludge reactor (UMSR) with falling water and reflux was constructed to remove nitrogen from anaerobic digestion of municipal wastewater with low C/N ratio. The bubble-free aeration mode in the UMSR reduces the disturbance to granular sludge. The main objectives of this study were to start-up UMSR, evaluate its pollutant removal performance, and gain a comprehensive insight into the biological mechanisms and conversions for simultaneous removal of carbon and nitrogen.

2. Methods

2.1. Experimental set up

The UMSR consisted of a 0.5 m high plexiglass column with an effective volume of 3.9 L. The bottom of the cylinder was connected to a 0.5 L cone (Fig. 1). The wastewater was fed into the reactor by a peristaltic pump. The top of the UMSR was open and the rising water could overflow from the top opening and fall into the reoxygenation water reflux tank around the column. The reoxygenation process was realized by falling water. A part of the reoxygenated water was refluxed into the UMSR and the remainder was discharged through the settling tank. The concentration of DO in the reactor was influenced by the reflux ratio.

2.2. Wastewater and seed sludge

The influent simulated the municipal wastewater after anaerobic digestion treatment and was synthetic wastewater of glucose with a COD concentration of 70 mg/L. The necessary nutrients supplied to

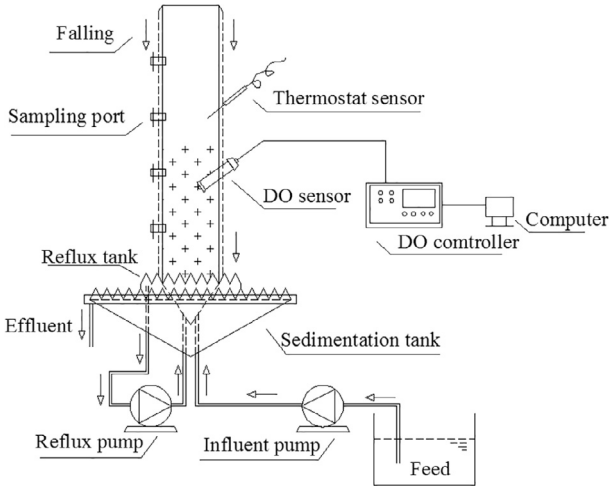


Fig. 1. Schematic representation of the lab-scale UMSR.

improve sludge growth during the acclimatization of inoculated sludge are as follows (mg/L): NH_4CI (70.0), KH_2PO_4 (19.7), Na_2SO_4 (14.2), $MgSO_4{\cdot}H_2O$ (39.0), $CaC1_2{\cdot}2H_2O$ (36.8), $CoC1_2{\cdot}6H_2O$ (1.2), $NiCl_2{\cdot}6H_2O$ (1.2), $Na_2Mo_4{\cdot}2H_2O$ (1.0), $FeC1_3$ (5.0), $CuSO_4{\cdot}5H_2O$ (5.0), $MnSO_4{\cdot}4H_2O$ (5.0), $ZnSO_4{\cdot}7H_2O$ (5.0), and $KHCO_3$ (1200).

The seed sludge for UMSR inoculation was activated sludge obtained from an anoxic tank with ongoing anaerobic/anoxic/oxic (AAO) process for treating domestic wastewater. The initial biomass in the UMSR in terms of mixed liquor suspended solids and mixed liquor volatile suspended solids were 4.66 and 1.29 g/L, respectively.

2.3. Analytical methods

Sampling from the influent and effluent of the UMSR reactor was conducted at an interval of two days. All samples were filtered with a 0.45 μm filter before analysis. TN, NH_4^+-N, NO_2^--N, and NO_3^--N were measured according to standard methods (APHA, 1998). DO, pH, and temperature were measured by oxygen, pH, and temperature meters (MT InPro6800, MT FE28, MT Company). Total organic carbon (TOC) and inorganic carbon (IC) were measured by a TOC analyzer (TOC-V CPH, Shimadzu Company).

2.4. Operation of the UMSR

First, the DO concentration was adjusted by regulating the reflux ratios at each stage. The DO in the reactor was detected by an on-line monitoring instrument which was also used to control the aeration of the reflux ratio. The DO values and reflux ratios for phases 1, 2, and 3 were 1.53 mg/L, 10:1, 0.37 mg/L, 4:1, and 0.5 mg/L, 5:1, respectively. The hydraulic retention time (HRT) and temperature in the reactor were maintained at 24 h and 25 ± 1 °C, respectively.

2.5. DNA extraction and Illumina MiSeq sequencing

The seed sludge, sludge during phase 2 (Day 60), and sludge during phase 3 (Day 128) were sampled and named S1, S2, and S3, respectively. The total DNA of the samples was extracted using the Bacteria DNA Isolation Kit Components (MOBIO) according to the manufacturer's instructions. Amplifications of bacterial 16S rRNA genes for the V3–V4 regions were performed with bacterial fused primers 341F and 805R (Herlemann et al., 2011; Hugerth et al., 2014). The specific procedure of polymerase chain reaction (PCR) was conducted according to a previous study (Zhou et al., 2015). The composition of PCR products of V4 region was determined by pyrosequencing using the Illumina Miseq sequencer platform (Shanghai Sangon, China).

2.6. Scanning electron microscopy observation

A sludge sample with a diameter of 0.5 mm was prepared by fixing the biomass with 2.5% glutaraldehyde with pH 6.8 at 4 °C for 1.5 h. The sample was washed three times with 0.1 mol/L phosphoric acid buffer at pH 6.8 for 10 min each time. The sample was dehydrated once each with 50%, 70%, 80%, and 90% ethanol and subsequently thrice with 100% ethanol for 10 min at each instance. Thereafter, the sample was rinsed for 15 min with a mixed liquor of ethanol and isoamyl acetate in a volume ratio of 1:1 and pure isoamyl acetate. The sludge sample was placed in the dryer for seven hours. Subsequently, the sample was pasted on the scanning electron microscope aluminum plate and sprayed with gold for one minute.

3. Results and discussion

3.1. Pollutant removal efficiency of the UMSR

To acclimate the denitrifying bacteria of different pathways, the inoculum was obtained from the anoxic tank in the AAO process because of the occurrence of denitrification in the traditional nitrogen removal process (Chen et al., 2016). According to the pollutant removal performance of the UMSR, the DO concentration was adjusted step by step by changing the reflux ratio. Additionally, the start-up period was divided into three phases with aerated water reflux ratios of 10:1, 4:1, and 5:1, respectively.

During phase 1, the influent NH_4^+-N concentration was 70 mg/L

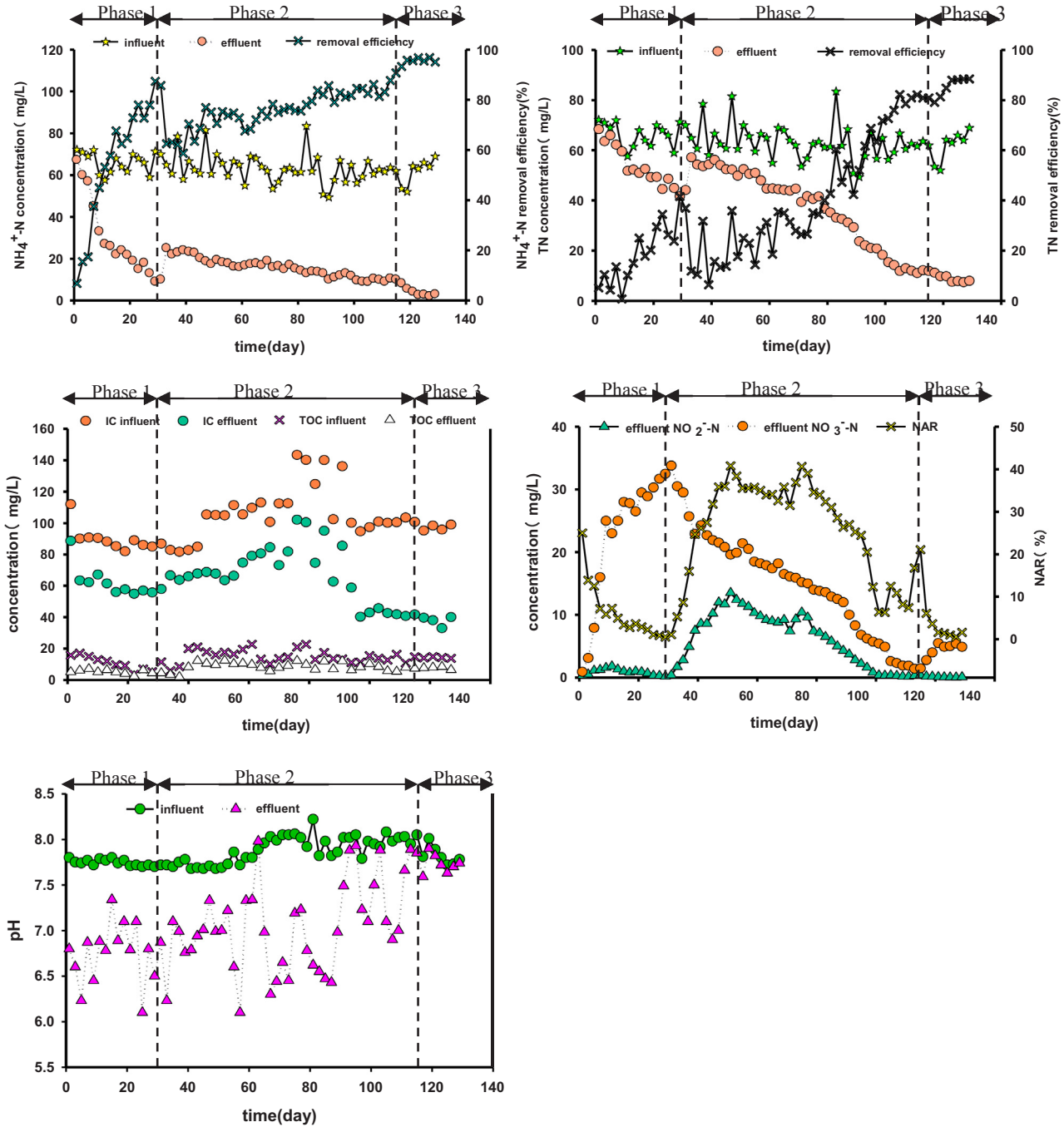


Fig. 2. Performance of UMSR in NH_4^+-N, TN, NOx^--N, TOC, IC and pH.

and the NH_4^+-N removal from the reactor effluent on the first day was relatively low. After 30 days (1st–30th days), the effluent NH_4^+-N was reduced to 10 mg/L (Fig. 2) and its removal rate reached 85%. After acclimation to the high DO in phase 1, the AOB in the seed sludge recovered activity and proliferated. At the beginning of phase 1, the concentrations of NO_2^--N and NO_3^--N were very low. However, after three days, the concentration of NO_3^--N increased sharply to 35 mg/L at the end of phase 1. A higher reflux ratio of 10:1 provided a high DO of about 1.53 mg/L in the reactor resulting in complete nitration process. Because of high DO and low TOC concentrations, the denitrification capacity of the sludge was insufficient and NO_3^--N could not be degraded, thus, resulting in high TN concentrations in the effluent.

In phase 2, when the reflux ratio was reduced to 4:1, the effluent NH_4^+-N sharply increased to 30 mg/L due to the lack of DO in the first 10 days of phase 2. Subsequently, it slowly decreased to 10 mg/L and reached a steady state on the 100th day. NO_2^--N slowly increased and then, fluctuated between 7 and 15 mg/L. Nitrite accumulation rate was found as 42% and the accumulation of nitrite provided a substrate for anaerobic ammonium oxidation (anammox) and enriched the pathways of denitrification (Du et al., 2016). Enrichment of anammox bacteria (AMX) and autotrophic denitrifying bacteria led to a decrease in NO_3^--N and an increase in the recovery of TN removal rate from 10% to 80% at the beginning of phase 2. In the latter part of phase 2, nitrate gradually decreased to 2 mg/L.

In phase 3, the increase of the reflux ratio allowed the AOB to utilise more DO, resulting in a further decrease in NH_4^+-N and a slight increase in NO_3^--N to 5 mg/L in the effluent. Finally, NH_4^+-N was reduced to less than 3 mg/L and the removal rate was above 95%. NO_2^--N and NO_3^--N were less than 1 and 5 mg/L, respectively. The total nitrogen removal rate was more than 90% and excellent nitrogen removal performance was obtained in the UMSR at 25 °C at an HRT of 24 h.

The removal rates of TOC and IC are closely related to the nitrogen removal performance. The removal rate of IC increased from 30% in phase 1 to 67% at the end of phase 3. As the removal rate of NH_4^+-N increased, the removal rate of IC correspondingly increased. The IC is used by autotrophic bacteria (AOB, NOB, and AMX) for growth and biosynthesis. The removal rate of TOC remained almost unchanged during the whole acclimation process.

The pH of the influent was higher than that of the effluent at the initial phase of start-up. With the extension of acclimation time and improvement of the denitrification capacity of the UMSR, the pH of the effluent gradually increased and was close to that of the influent during the later period of phase 3.

3.2. Diversity of the microbial community

To comprehend the effects of the sludge acclimation method on the performance of the UMSR, activated sludges S2 (60th day) and S3 (128th day) were sampled for DNA extraction and PCR amplification while the inoculated activated sludge (S1) was used as control. Reads and diversity indices of microbial communities in the inoculum and the acclimated sludges are illustrated in Table 1. Among the indices, Good's coverage of the three samples indicated that the sequence libraries constructed in the present research could cover the diversity of microbial communities in the sludge samples (Zhen et al., 2014). Chao 1 and ACE species richness index indicated that maximum number of species could be obtained when saturated sequencing was reached. Shannon index was used to characterize the diversity of the microbial community and it indicated results for species abundance and uniformity of species distribution.

As illustrated in Table 1, the Chao1, ACE, and Shannon indices for sludges from phases 2 and 3 were remarkably higher than those of the inoculum. The results indicate that species richness and diversity of microbial communities had been remarkably enhanced. The change in diversity was a response of the microbial community to adapt to environmental changes to better metabolize the pollutants. The more complex microbial community structure of the UMSR suggests that the ecological self-balancing capacity of the reactor gradually enhanced with increasing acclimation time (Fig. 2). Compared with the inoculum in which heterotrophic denitrification was the main metabolic process, the diversity of acclimated microbial community increased because of occurrence of simultaneous partial nitrification, anaerobic ammonium oxidation and denitrification (SNAD) and carbon removal in the single-stage UMSR. As illustrated in Fig. 2, the pollutant removal in phase 3 was significantly better than that in the other phases.

3.3. Microbial community structure

Figs. 3 and 4 show the microbial community structure of the three samples at phylum, class, and genus levels. The phylum level results (Fig. 3a) demonstrate that the microbial community structure in phase 3 was obviously different from that in the inoculum after acclimation. The proportions of the main phyla Proteobacteria, Planctomycetes, Bacteroidetes, Chloroflexi, and Acidobacteria were 45.31%, 1.11%, 31.07%, 4.08%, and 1.34% in S1, 37.55%, 9.02%, 8.15%, 14.63%, and 13.32% in S2, and 32.83%, 18.28%, 1.95%, 11.3%, and 10.92% in S3, respectively (Fig. 3a). It is worth noting that these phyla are critical for biological nitrogen removal during wastewater treatment. It was found that the proportion of Planctomycetes in S3 increased significantly from 1.11% to 18.28% in comparison to S1 due to long-term acclimation to low DO and low C/N in the influent. This was because some species of Planctomycetes displayed the ability to grow anaerobically and autotrophically via oxidation of ammonium (Fuerst, 2004). There was also a marked increase in the proportions of Chloroflexi, Acidobacteria, and Nitrospirae. Although the proportion of Proteobacteria decreased from 45.31% to 32.83%, it was still the highest in S3. Thus, Proteobacteria still played a key role in the biodegradation process of the microaerobic wastewater treatment system (Yang et al., 2014a). A previous study has revealed that one of the most dominant aerobic ammonium oxidizing bacteria (AerAOB) inoxygen-limiting process were Proteobacteria (Liu et al., 2012a). Thus, the large proportion of Proteobacteria present in the reactors contributed to the partial nitrification process. The α-Proteobacteria, β-Proteobacteria, γ-Proteobacteria, or δ-Proteobacteria were showed the highest proportions in each of the three sludge samples (Fig. 3b). In general, the α-proteobacteria carried out a wide variety of metabolic reactions such as photosynthesis, nitrogen fixation, ammonia oxidation, and methylotrophy. In contrast to other studies on the treatment of wastewater with low COD/TN ratio by the microaerobic process, the proportion of Bacteroidetes decreased dramatically from 31.07% to 1.95%.

To reveal the biological mechanism of nitrogen removal in the

Table 1
Diversity of community in inoculum and activated sludge from phases 2 and 3.

Phase	Sample	Reads	3% Distance				
		Raw reads	OTUs	Chao1 richness estimation	ACE index	Shanno diversity index	Good's coverage
Inoculum	S1	23,346	874	1240	1378	5.05	0.98
2 (60 day)	S2	62,215	2778	3379.74	3591.38	6.02	0.98
3 (128 day)	S3	57,381	2831	3510.23	3709.29	6.32	0.98

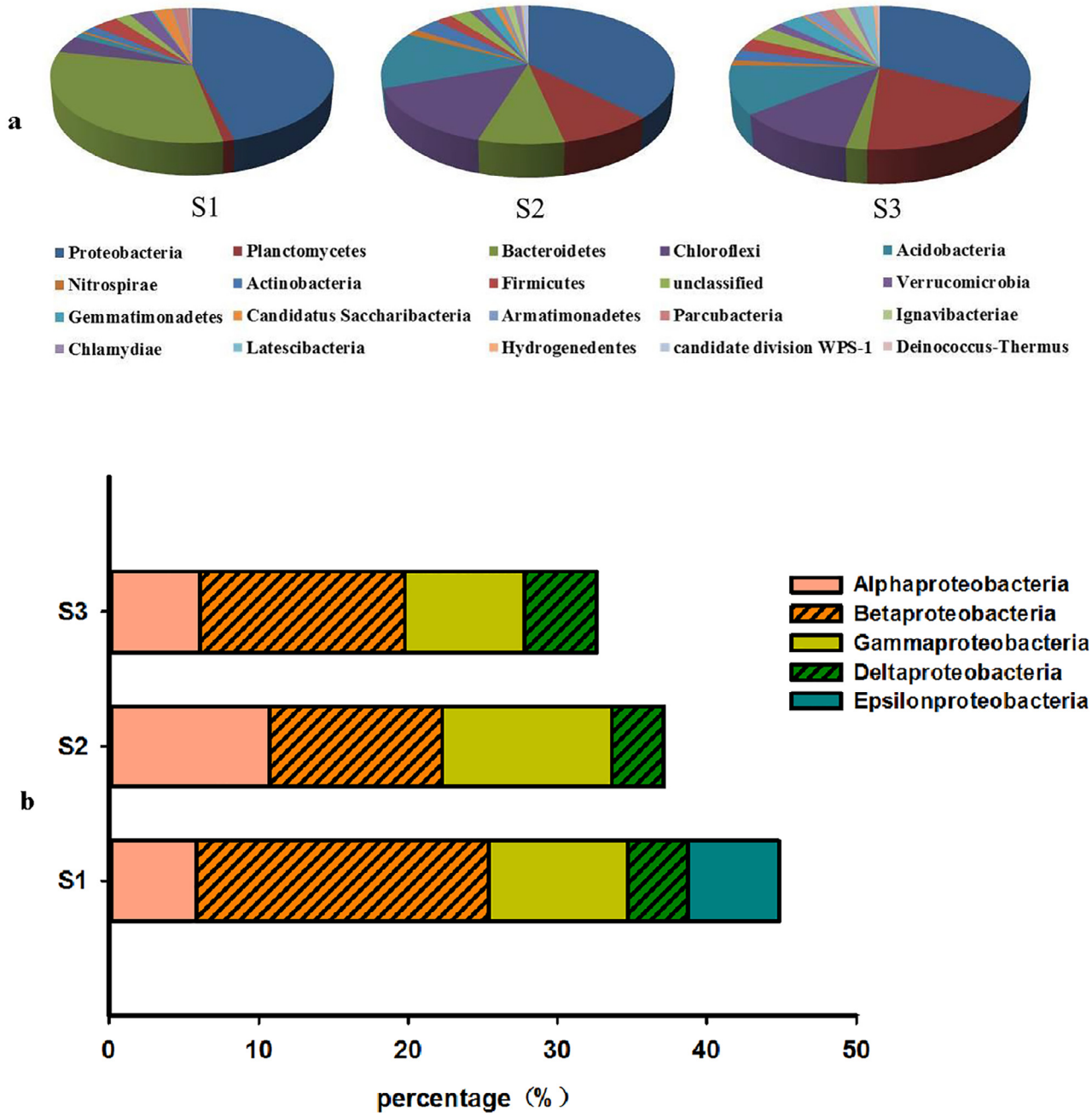


Fig. 3. Bacterial community structure at phylum and class in Proteobacterial of three samples.

UMSR, further analysis of the microbial communities at the genus level was carried out (Fig. 4). The AOB (*Nitrosospira*) were only about 0.25% in S1 because the inoculum was anoxic activated sludge obtained from the AAO process. During the process of acclimation, the abundances and diversity of ammonium oxidation-related bacteria gradually enhanced. The main species of AOB increased from just *Nitrosospira* to *Sphingomonas*, *Nitrosomonas*, and *Nitrosospira* with respective proportions of 0.93%, 0.35%, and 0.01% in the S3. This was different from the AAO process where *Nitrosospira* was the main species of AOB (Zhang et al., 2017). In this study, the genera related to anaerobic ammonium oxidizing bacteria (AnAOB) were detected as the predominant functional microbes in S2 and S3 and the proportions of *Candidatus Kuenenia*, *Candidatus Brocadia*, *Candidatus Scalindua* and *Candidatus Anammoxoglobus* in S3 were 0.37%, 0.22%, 0.08%, and 0.03%, respectively (Cabello et al., 2009). Compared to the proportion of AerAOB (1.29%), the proportion of AnAOB was 0.70%, and the enrichment of these genera contributed to excellent ammonium removal in the UMSR. This was consistent with previous phylogenetic studies which indicate that α- Proteobacteria and β-Proteobacteria related AerAOB and Planctomycetales-like AnAOB were the predominant bacteria in the CANON system (Liu et al., 2012b). AMX needed only ammonium, nitrite, and CO_2 as the major growth substrates and nutrients (Shively et al., 2009). The structure of activated sludge granules and floc not only formed the DO gradient but also produced a nutrient gradient. Camp et al., (2007) found that the system of anammox was anoxic and characterized by dynamic nutrient gradients.

Compared to S1, the diversity and proportion of AOB-related bacteria in the UMSR increased. In S1, the proportion of NOB (0.5%) was greater than that of AOB (0.25%), while it was similar (1.32%) to that of AOB (1.29%) after acclimation to low DO. The main mechanisms for controlling NOB growth in the reactors were substrate gradient and competition for oxygen by AOB and NOB (Brockmann and Morgenroth,

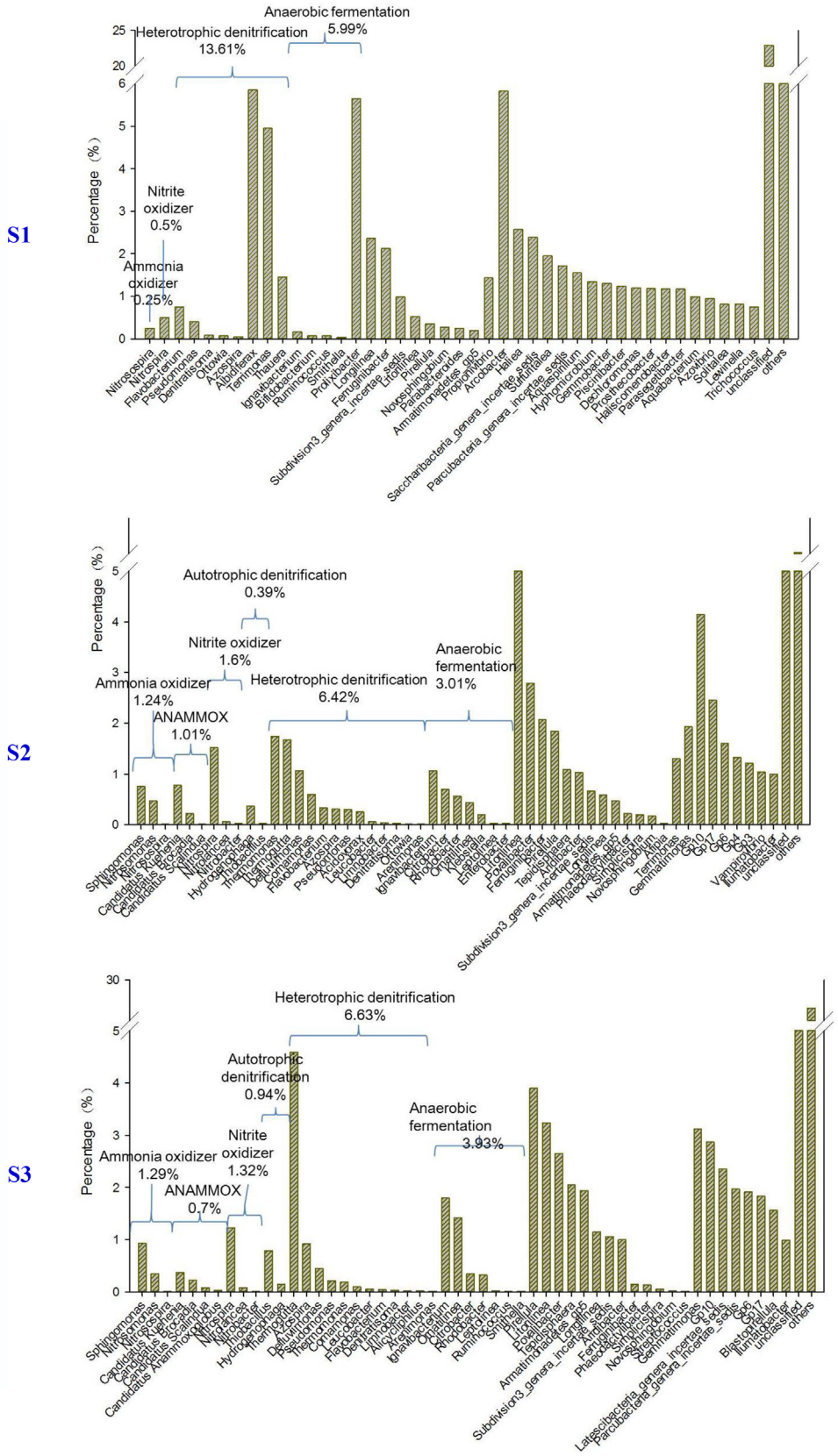


Fig. 4. Bacterial community structure at genus of three samples.

2010; Pérez et al., 2015). In addition, during phases of acclimation, the system had the selective advantage of *Nitrospira*-like NOB over *Nitrobacter*-like NOB because the half-saturation constants (Km) for *Nitrospira* were between 9 and 27 μM nitrite which were less than those of *Nitrobacter* with Km values between 49 and 544 μM nitrite. Additionally, *Nitrospira* bacteria were adapted to live under significant

Table 2
Nitrogen removal efficiency and active microbial fraction.

	Reflux ratio	DO (mg/L)	sampling time	Effluent (mg/L)			Removal rate (%)			Proportion of functional bacteria (%)				
				NH_4^+-N	NO_2^--N	NO_3^--N	NH_4^+-N	TN	IC	AOB	NOB	AMX	Autotrophic denitrification	Heterotrophic denitrification
Phase 1	10:1	1.53	–	10	0	35	85	40	30	–	–	–	–	–
Phase 2	4:1	0.37	60	20	10	20	70	40	38	1.24	1.60	1.01	0.39	6.42
Phase 3	5:1	0.50	128	3	0	5	95	90	67	1.29	1.32	0.70	0.94	6.63

substrate limitations (Nowka et al., 2015).

During the start-up, a high proportion of NOB was enriched at the end of phase 1 because of high DO. However, the autotrophic denitrifying bacteria and AMX were not enriched, resulting in NO_3^--N in the effluent reaching a concentration of 35 mg/L (Table 2). The low DO ranging between 0.35 and 0.50 mg/L in phases 2 and 3 inhibited the growth and activity of NOB. Although the DO in phase 3 was slightly higher than that in phase 2, the DO concentration of 0.5 mg/L in phase 3 still could not meet the oxygen affinity of NOB (Laanbroek and Gerards, 1993). Therefore, the proportion of NOB for both phases (2–3) gradually reduced from 1.60% to 1.32%. However, the saturation constant of DO for AOB was as low as 0.2–0.4 mg/L (Laanbroek and Gerards, 1993), therefore, the proportion of AOB increased from 0.25% in the inoculum to 1.24% in phase 2, and 1.29% in phase 3. As a result, the removal rate of ammonium gradually increased from 60% in phase 2 to 90% in phase 3. The increase in the proportion of AOB resulted in the accumulation of nitrite which provided substrates for various subsequent denitrification pathways. Heterotrophic denitrification was the most important process for nitrogen removal in the inoculum because of the proportion of heterotrophic denitrifiers in the microbial community (Fig. 4). After sludge acclimation, the proportion of heterotrophic denitrifiers decreased from 13.61% to 6.63%. In addition, the proportion of heterotrophic denitrifying bacteria in the long-term low C/N ratio influent was balanced, and the proportions in S2 and S3 were similar. Under the condition of low C/N ratio in the wastewater, autotrophic denitrifying bacteria which could denitrify with no supplement of carbon source (Claus and Kutzner, 1985) were detected to increase from 0% in the inoculum to 0.94% in phase 3, and the increase in the proportion of autotrophic denitrifying bacteria played a key role in denitrification in the UMSR resulting in a decrease of NO_3^--N in the effluent from 35 mg/L to 5 mg/L. As mentioned previously, anammox was another nitrogen removal approach which needed no carbon source (Chamchoi et al., 2008). The enrichment of autotrophic denitrifying bacteria contributed to excellent TN removal rate.

The long-term acclimation to low C/N ratio and low DO strengthened the role of AOB and autotrophic denitrifying bacteria while the microaerobic conditions created a DO gradient in the activated sludge granules and flocs to allow aerobes, anaerobes, and facultative aerobes, including AOB, NOB, AMX, autotrophic denitrifying bacteria, and heterotrophic bacteria, to thrive in different depths of the granules and flocs. The various functional microorganisms could cooperate to remove pollutants in the UMSR, especially for nitrogen removal from wastewater with a low COD/TN ratio, resulting in removal rates of 95% for ammonia nitrogen and 90% for TN. The oxygen for the novel UMSR was supplied by recirculating falling effluent instead of aeration. The supplied oxygen could meet the demands of nitrogen removal in the reactor. Therefore, the energy consumption for treating wastewater with low COD/TN ratio using this novel reactor will be significantly reduced. The economic feasibility will be further discussed in future research.

3.4. Characteristics of sludge granules

Granular sludge was found to occur during the process of sludge acclimation. The granulation process of microaerobic sludge included three stages: seed sludge acclimation, flocculation, and granulation. The SEM images of granulation process were shown in E-supplementary data and could be found in e-version of this paper online. After 30 days of acclimation, irregularly shaped inoculum gathered regularly in groups. A large volume of filamentous bacteria and extracellular polymeric substances (EPS) could be observed. The denitrifying sludge was characterized by self-aggregation (Green et al., 1994) because denitrifying bacteria, such as heterotrophic bacteria, could produce high quantities extracellular polysaccharides. Microaerobic granules were aggregates of various minerals, inorganic compounds, EPS, and microorganisms (Adav and Lee, 2008). Additionally, EPS promoted fine particle debris to form granules. (Chen and Lee, 2015) claimed that *Brevundimonas*, whose proportion in the UMSR sludge samples was detected as 0.4%, is an effective EPS producer. It has been reported that *Pseudomonas*, *Clostridium*, *Thauera*, and *Arthrobacter* produce a large number of viscous substances, such as excess alginate-like extracellular polysaccharide and extracellular c-di-GMP, which are precursors of granules (Yang et al., 2014b). These bacteria were detected in the sludge samples. In an environment with low DO concentrations, the filamentous bacteria could gain more DO and had growth advantages because of increased specific surface area of their extended mycelium which formed aggregate structures for sludge granules.

Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.biortech.2018.08.016.

Granulation of sludge occurred in the system within 50 days, however, the size of granules was small. After acclimation for 100 days, larger and denser granules with an average diameter of 200 um were observed. The formed granular sludge was regular and compact in appearance and gaps in the surface layer and the internal layer the granular sludge existed for transporting internal and external substances. This was due to continuous production and release of gas from the sludge granules by denitrifying bacteria and nammox bacteria (Tang et al., 2011). The microaerobic system exhibited increased nitrogen removal as the size and structure of formed granules became bigger and denser which is in agreement with the results of a previous study (Huang et al., 2015).

In the internal structure of microaerobic granules, heterotrophs used COD as a substrate to secrete excess EPS and autotrophic organisms formed hydrophobic cell surfaces which formed dense anaerobic cores and outer layers through the cluster-cluster aggregation mechanism (Cui et al., 2014; Tijani et al., 2015). There were abundant microorganisms in the granules which were formed by polymerization of a large number of spherical, bacillus, and filamentous bacteria, and the bacilli were attached to the surface of granules or flocs. AOB were mostly rod-shaped, some heterotrophic bacteria were filamentous, and AMX were mostly spherical. The varied microbial community structure was the premise of rapid transformation and degradation of pollutants. Moreover, the staggered lifestyle of the microbial community was caused by nutrient transmission and strain reproduction which was conducive to division of labour and cooperation among the microbial community as well as efficient removal of pollutants.

Rathnayake et al., (2015) also found granular processes for partial nitrification and denitrification. Partial nitrification granules of AOB dominated by *Nitrosomonas* were successfully cultured (Liang et al., 2015). A partial nitrification aerobic granular reactor was used to treat

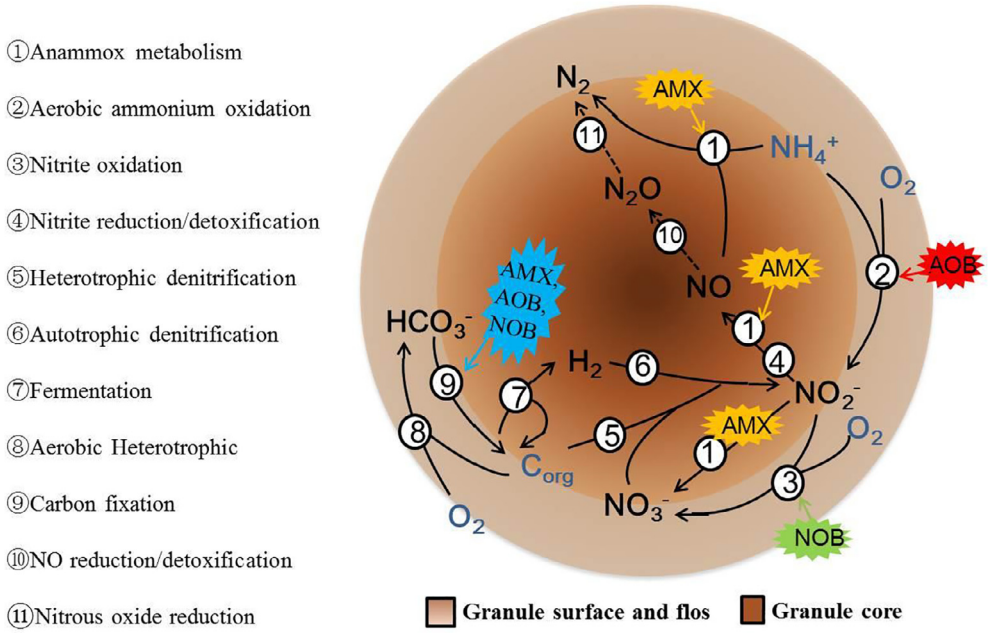


Fig. 5. Schematic overview of nitrogen conversions in UMSR. Substances indicated by blue are supplied to granule sludge or flocs through inflow and backflow, respectively. On the surfaces of particles and flocs, organic carbon (C-org) is oxidized to carbon dioxide (CO_2) by heterotrophic bacteria and subsequently fixed by autotrophic organisms in the system. NH_4^+ is oxidized to nitrite (NO_2^-) by AOB. A part of the generated NO_2^- is further oxidized to NO_3^-. Hydrogen (H_2) can be formed by anaerobic fermentation of organic carbon. Heterotrophic or autotrophic denitrifying bacteria use C-org or molecular hydrogen as electron donors to reduce NO_3^- formed in the particles to NO_2^- in the anaerobic granule core. NO_2^- is further reduced to nitric oxide (NO) and NO is converted to nitrogen dioxide gas (N_2) by autotrophic denitrifying bacteria or is combined with NH_4^+ by AMX. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

highly saline and ammonia laden wastewaters (Wan et al., 2014a; Wan et al., 2014b). Balanced growth of AOB and NOB was successfully achieved in cultured granules and the formation of granule sludge improved the ability of the system to remove pollutants.

3.5. Integrating inferred niches and activities in a granule model

An ecological model proposed by (Speth et al., 2016) offers an explanation for the effective performance of the SNAD reactor. Based on the functions derived from the bacterial community structure at the genus level, the ecological model, catalysed by the microbial biomass, is suitable for the nitrogen cycle reactions of this study (Fig. 5). Because the outer layer of the flocculent biomass or granule surface is exposed to oxygen, the main processes in the flocculent biomass or granule surface that are dominated by heterotrophic bacteria and nitrifiers (AOB and NOB) are aerobic oxidation of organic carbon to CO_2 and complete nitrification of ammonium to nitrate. The formed CO_2 either escapes to the atmosphere or is used by the primary producers (AOB and NOB) for growth and biosynthesis. Inside the particles, a part of nitrite is used by AMX to oxidize ammonium and produce nitrate. Based on the stoichiometry of the anammox process, one mole of nitrite reduced produces 0.3 mol of nitrate resulting in accumulation of nitrate in the effluent (Fig. 2). Thus, excess nitrate accumulation in the system might be mitigated by the presence of (partial) denitrifiers that catalyse the conversion of nitrate to nitrite.

A part of the formed nitrate can be reduced to nitrite using either organic matter or hydrogen as the electron donor by heterotrophic and autotrophic denitrifying bacteria, making additional nitrite available for AMX. The hydrogen required for autotrophic nitrate reduction can be formed through fermentation of organic matter by anaerobic fermentation bacteria. The organic matter required for both nitrate reduction and fermentation can come from the influent, dead organic matter in the granule, or extracellular granule matrix synthesized by autotrophs. A part of the formed nitrite during autotrophic and heterotrophic denitrification converts to N_2. AMX might metabolize the NO to N_2 (Kartal et al., 2010a,b).

4. Conclusions

To conduct municipal wastewater with low C/N ratio after anaerobic digestion, a novel UMSR was constructed and its performance was investigated at a temperature of 25 °C and an HRT of 24 h. Ammonium and TN removal achieved in the UMSR were as high as 92.35% and 90.41%, respectively. Formation of particle sludge provided DO and pollutant concentration gradients which were conducive to division of labour and cooperation among microbial communities as well as efficient removal of pollutants. The inferred niches and activities of the microbial communities in the granule model offered an explanation for the effective performance of the UMSR.

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